A Project Report on

Smart Collabration And Career Guidance System

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Degree

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CSE(Artificial Intelligence and Machine Learning)

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DECLARATION

We declare that this project work is composed by ourselves, that the work contained herein is our own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

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ABSTRACT

Students and professionals today face significant challenges in navigating complex career landscapes, finding quality mentorship, and engaging in effective collaborative learning. Traditional educational platforms often operate in silos, lacking personalized career advice, real-time interactive spaces, and direct access to industry experts. These disjointed approaches frequently lead to skill gaps, uninformed career choices, and a lack of consistent academic motivation.This project proposes a Smart Collaboration and Career Guidance System designed to bridge the gap between academic learning and professional development. The platform integrates an intelligent AI-driven Career Bot that provides personalized roadmap recommendations, answers career-related queries, and guides users based on current industry trends. To foster direct professional growth, it incorporates an interactive mentorship module where students can find, schedule, and connect with experienced mentors, supported by a secure integrated payment mechanism.To enhance peer-to-peer learning, the system features dynamic study spaces that allow users to create and join topic-specific collaboration rooms for shared studying and knowledge exchange. Deployed as a comprehensive, interactive web-based platform, the application provides a seamless user experience focused on community, collaboration, and clear progression. By combining intelligent virtual career counselling, expert mentorship access, and real-time collaborative environments, the proposed platform transforms traditional, isolated learning models into a highly connected and user-centred ecosystem, empowering students to make informed decisions and build robust skill sets.

Keywords: Career Guidance, AI Chatbot, Collaborative Learning, Mentorship Platform, Interactive Study Rooms, Professional Development

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CHAPTER 1
INTRODUCTION

Effective time management is essential for academic success, yet many students struggle to create realistic and sustainable study schedules. Conventional planners and digital calendar applications rely primarily on deadlines and priority levels, offering static scheduling without considering individual productivity patterns, subject-specific weaknesses, stress levels, or peak performance hours. Such rigid approaches often lead to inconsistent performance, inefficient study allocation, and academic burnout.

Advancements in Artificial Intelligence enable the development of adaptive systems capable of learning from user behaviour and personalizing decision-making processes. An intelligent study planner should analyse completion history, productivity trends, stress indicators, and biological rhythm patterns to optimize workload distribution. Incorporating wellness-aware mechanisms is equally critical to ensure that performance improvement does not compromise mental health and long-term sustainability.

The proposed AI-Based Adaptive Study Planner generates personalized, self-optimizing schedules using behavioural analytics and predictive performance metrics such as productivity score, subject weakness score, and schedule confidence score. The system dynamically adjusts workload based on user consistency, detects peak productivity periods through biological rhythm analysis, integrates stress-aware scheduling and structured focus techniques, and provides explainable guidance to enhance transparency and engagement. By combining adaptive intelligence with wellness-centered optimization, the system transforms traditional static planning into a personalized, predictive, and sustainable academic assistance framework.

1.1 BACKGROUND OF THE STUDY

The twenty-first century has witnessed an unprecedented transformation in the domain of education, driven by the rapid proliferation of digital technologies, the internet, and artificial intelligence. Traditional modes of academic learning — centred around classroom instruction, printed textbooks, and one-on-one teacher interaction — are gradually giving way to technology-mediated learning environments that offer greater flexibility, accessibility, and personalisation. This paradigm shift has opened new avenues for students to learn collaboratively, seek expert guidance remotely, and plan their academic and professional futures with greater precision and confidence.

In the Indian educational ecosystem, students navigate a series of high-stakes academic transitions that fundamentally shape their career trajectories. The first major transition occurs at the Class 10 level, where students must select a stream of study — Science (MPC or BiPC), Commerce, or Arts/Humanities. The second transition occurs after Class 12 (Intermediate), where students must choose an appropriate undergraduate programme and institution. The third and perhaps most challenging transition occurs during the undergraduate degree years, when students must determine whether to pursue employment, higher education, or entrepreneurship. At each of these critical junctures, students are expected to make well-informed, life-defining decisions — yet the majority do so without adequate access to professional guidance, structured mentorship, or reliable information.

According to various educational surveys and reports published by the National Sample Survey Office (NSSO) and All India Council for Technical Education (AICTE), a significant proportion of engineering and degree graduates in India are underqualified or misaligned with industry requirements. A key contributing factor is the absence of structured career guidance during secondary and higher secondary education. Students frequently select courses based on peer pressure, parental expectation, or incomplete understanding of career prospects, rather than informed analysis of their interests, aptitudes, and market demand.

Simultaneously, collaborative learning has been recognised globally as one of the most effective pedagogical strategies for improving academic performance and motivation. Research in educational psychology consistently demonstrates that students who engage in peer collaboration, structured discussion, and group problem-solving develop stronger conceptual understanding, better communication skills, and greater academic resilience compared to those who study in isolation. Despite this, most Indian students lack access to digital platforms specifically designed to facilitate academic collaboration — particularly those that integrate communication, task management, and mentorship within a unified environment.

The COVID-19 pandemic (2020–2022) further accelerated the adoption of online learning tools but simultaneously exposed the severe limitations of existing platforms. Tools such as Google Classroom, Microsoft Teams, and Zoom were adopted widely, yet they serve only narrow purposes — file sharing and video conferencing — and lack intelligent advisory features, structured mentor-student matching, or integrated financial mechanisms to support professional mentorship. Students who require expert guidance must navigate fragmented

ecosystems involving separate communication apps, third-party payment platforms, and unstructured social media groups.

Artificial Intelligence (AI) has emerged as a transformative force in modern education. AI-powered systems can analyse large volumes of educational and behavioural data to generate personalised recommendations, predict student performance, and deliver context-aware guidance at scale. Notably, Retrieval-Augmented Generation (RAG) — a technique that combines structured information retrieval from a curated knowledge base with the generative capabilities of large language models (LLMs) — has proven highly effective in building domain-specific intelligent assistants. When applied to career guidance, a RAG-based chatbot can retrieve reliable, structured information about streams, courses, and career paths and generate student-friendly, context-specific responses in natural language.

In response to these challenges and opportunities, the present project proposes and develops **CollabSphere**, a Smart Collaboration and Career Guidance System. CollabSphere is a comprehensive, full-stack web platform that integrates four major capabilities into a single, cohesive environment: (1) subject-based collaborative study rooms with real-time chat and task management, (2) an AI-powered career guidance chatbot built on a RAG pipeline using the Groq-hosted Llama 3.1 language model, (3) a structured mentor-student connection system with role-based dashboards, and (4) a Razorpay-integrated payment mechanism for booking professional mentorship sessions. The system is designed to serve students across all academic levels — from Class 10 through undergraduate degree — and is accessible through any modern web browser without requiring any software installation.

CollabSphere draws upon established principles from computer-supported collaborative learning (CSCL), human-computer interaction (HCI), and educational data science, and applies modern web technologies including React, Flask, SQLAlchemy, PostgreSQL, and cloud deployment via Vercel to deliver a robust, scalable, and user-centred academic platform.

1.2 PROBLEM STATEMENT

Despite the widespread availability of digital tools and internet connectivity, the Indian student community continues to face a set of persistent and interrelated challenges that significantly impede effective learning, informed career decision-making, and equitable access to expert mentorship. The following problem areas have been identified through observation, academic literature, and analysis of existing platforms:

1. Absence of a Unified Collaborative Learning Environment: Students who wish to study collaboratively with peers are compelled to use a fragmented combination of tools — WhatsApp for communication, Google Docs for shared notes, Trello or Notion for task tracking, and Zoom for meetings. None of these platforms are specific to academic study groups, nor do they integrate all necessary functions into a single environment. There is no dedicated, student-centric platform that allows users to create subject-specific study rooms, track shared tasks on a structured board, communicate in group chats, and invite mentors — all within the same application.

2. Lack of Personalised and Reliable Career Guidance: A substantial proportion of students, particularly at the Class 10 and Class 12 levels, make critical academic decisions without access to credible, personalised career guidance. The available resources are either overly generic (pamphlets, general websites), commercially biased (coaching centre promotions), or socially influenced (peer advice, family preference). There exists no intelligent, readily accessible system that can analyse a student's interests and background to deliver structured, evidence-based career recommendations covering streams, courses, entrance examinations, skill requirements, and long-term career paths.

3. Inaccessibility of Professional Mentors: Quality mentorship is disproportionately available to students in metropolitan areas and private institutions, while those in smaller cities, rural areas, and government colleges face significant barriers to accessing experienced professionals. Furthermore, even where mentors are available, there is no structured, transparent, and affordable mechanism for students to discover appropriate mentors, assess their expertise and charges, initiate connection requests, and conduct financial transactions in a secure and organised manner.

4. Non-Existence of an AI-Driven Career Advisory Tool for Indian Students: Existing AI-based chatbots (e.g., general-purpose models like ChatGPT) are not grounded in structured, domain-specific, and Indian-context career knowledge. Their responses may be generic, inconsistent, or culturally inapplicable. There is a clear need for an AI chatbot specifically trained or augmented with a structured career knowledge base covering Indian academic streams, university programmes, entrance examinations, and industry career paths — capable of providing reliable, multi-turn, context-aware conversations with students.

5. No Transparent Payment and Mentorship Booking Mechanism: Even when students and mentors agree to engage, the financial transaction layer is typically handled informally or

through third-party platforms, lacking transparency, security, or accountability. There is no integrated system that allows students to view a mentor's per-minute charges, book a session, make a verified digital payment, and have the funds released to the mentor upon session completion — with full transaction history maintained for both parties.

6. Absence of Role-Based, Dedicated Dashboards for Students and Mentors: Existing platforms do not differentiate meaningfully between the needs of students and mentors. Students need features for joining rooms, seeking guidance, and booking mentors; mentors need tools to manage connection requests, track assigned students, monitor earnings, and onboard into payment systems. The absence of role-specific interfaces increases cognitive load and reduces platform usability significantly.

These six problem areas collectively constitute a complex, multi-dimensional challenge that no single existing platform adequately addresses. CollabSphere is designed specifically to resolve all of the above challenges through a unified, intelligent, and role-aware web application.

1.3 OBJECTIVES OF THE PROJECT

The primary objective of this project is to design, implement, and deploy **CollabSphere**, an integrated Smart Collaboration and Career Guidance System that provides students and professional mentors with a unified, AI-augmented digital platform for collaborative learning, informed career planning, and structured mentorship. The specific objectives of the project are as follows:

1. To design and implement a secure, role-based user authentication and account management system that supports two distinct user roles — Student and Mentor — with JWT (JSON Web Token)-based session management, password hashing using Werkzeug's bcrypt algorithm, token-based email password recovery via a 15-minute expiry reset link, and profile management (bio, interests, phone number, per-minute charges for mentors).

2. To build subject-specific collaborative study rooms that enable students to create rooms with a subject title and description, send and manage join requests (status: pending / approved / rejected), assign roles to members (admin / member / mentor), and conduct persistent group chat communication — all managed through a secure REST API with JWT-protected endpoints.

3. To implement a Kanban-style task management system within each study room, enabling room members to create tasks with title, description, and deadline, assign tasks to specific

members, and track task progress across three status columns: Todo, In Progress, and Done — facilitating structured, team-based academic project management.

4. To develop an AI-powered career guidance chatbot using a Retrieval-Augmented Generation (RAG) pipeline that (a) retrieves the most relevant passages from a structured career knowledge base (career_docs.txt) using TF-IDF scoring, (b) augments the retrieved context into the system prompt of the Groq-hosted Llama 3.1-8b-Instant language model, and (c) maintains multi-turn conversation history (last 6 turns) to generate context-aware, student-friendly career guidance covering stream selection, course recommendations, skill roadmaps, internship suggestions, and certification guidance for students from Class 10 through degree level.

5. To create a structured mentor discovery and connection system that allows students to browse available mentors with full profile details (bio, interests, per-minute charges, discount percentage), send personalised connection requests with an optional message, and track the status (pending / approved / rejected) of their requests — while enabling mentors to review, approve, or reject incoming connection requests through a dedicated dashboard.

6. To integrate a Razorpay-based payment gateway for enabling secure, per-minute mentorship bookings — including order creation, Razorpay checkout integration, cryptographic payment signature verification, mentor onboarding via Razorpay Route for direct payouts, an escrow-based payment release system, webhook handling for real-time payment event processing, and complete payment history accessible to both students and mentors.

7. To design and implement role-specific interactive dashboards for both Students and Mentors — where the Student Dashboard provides tabs for Overview, Find Mentor, Connections, Rooms, Career Bot, Courses, and Profile; and the Mentor Dashboard provides tools for managing student connections, tracking active sessions, monitoring earnings, and completing Razorpay onboarding.

8. To deploy the application as a production-ready system on cloud infrastructure, with the React + Vite frontend hosted on Vercel and the Flask REST API backend hosted with PostgreSQL as the primary relational database, supported by environment-based configuration management via .env files and CORS-enabled cross-origin access.

9. To evaluate the functional correctness and usability of the system through structured test cases covering authentication, room management, task operations, API responses, chatbot accuracy, payment flows, and cross-browser compatibility.

1.4 SCOPE OF THE PROJECT

The scope of the CollabSphere — Smart Collaboration and Career Guidance System — is defined by the boundaries of its intended functionality, target user base, technological environment, and the limitations of the current implementation. The system is designed as a multi-user, role-based, web-accessible academic platform serving students and professional mentors in the Indian educational context.

1.4.1 Functional Scope

The following functional areas are included within the scope of the current implementation:

User Authentication and Profile Management: The system supports complete user lifecycle management including registration with role selection (Student or Mentor), secure login with JWT token issuance, token-protected API access, password reset via time-limited email links, and full profile editing (name, bio, phone number, interests, per-minute charge for mentors, and Razorpay account details).

Collaborative Study Rooms: Students can create subject-specific rooms, invite peers, manage join requests, assign roles, and maintain room membership. Each room serves as a self-contained workspace for group collaboration.

Task Management (Kanban Board): Each room supports a shared task board with three columns (Todo / In Progress / Done). Tasks can be created, assigned to room members, and updated. This facilitates structured project and study management within student groups.

Real-Time Group Chat: Each room includes a persistent group messaging feature. All messages are stored in the database and retrieved on room access, providing a shared communication history for room members.

AI Career Guidance Chatbot: The chatbot covers career guidance for students from Class 10 through undergraduate degree level. The knowledge base includes stream selection, course choices, skill development paths, higher education options, entrance examinations, and career domains. The chatbot supports multi-turn conversations and falls back to general LLM knowledge when the local knowledge base does not have sufficient coverage.

Mentor Discovery and Connection: Students can browse all registered mentors, view profiles and rates, and send connection requests. Mentors can approve or reject requests. Both parties can track connection status from their respective dashboards.

Payment and Mentorship Booking: The payment scope includes Razorpay order creation, front-end checkout integration, signature verification, mentor onboarding via Razorpay Route, payment history, webhook processing, and payment release. The system supports per-minute billing and discount application.

Notifications: Automated transactional emails are sent for password reset. These are delivered through a notification service integrated into the backend.

1.4.2 Technical Scope

The system is built as a two-tier web application:

- **Frontend**: React 19 with Vite, React Router DOM, Axios, Tailwind CSS, Lucide React icons, Firebase for client-side auth integration — deployed on Vercel.
- **Backend**: Python Flask REST API with SQLAlchemy ORM, JWT Extended, PostgreSQL (production) / SQLite (development fallback), Razorpay SDK, Groq API, Firebase Admin SDK — deployed on a cloud server with Gunicorn as the WSGI server.

The system is designed to be accessible through any modern web browser (Chrome, Firefox, Edge, Safari) without requiring software installation on the client side.

1.4.3 Out of Scope

The following features are explicitly outside the scope of the current implementation and are identified as potential future enhancements:

- **Live audio/video conferencing**: WebRTC-based real-time video or voice calls between students and mentors are not implemented in the current version.
- **Mobile applications**: Native Android or iOS applications are not developed; the platform is a responsive web application only.
- **LMS integrations**: Integration with third-party Learning Management Systems (e.g., Moodle, Canvas, Google Classroom) is not included.

- **Automated AI mentor matching**: The current system provides manual browsing; algorithmic matching of students to mentors based on compatibility scores is a future enhancement.
- **Super-admin platform dashboard**: A centralised administrative panel for platform-level user management, analytics, and content moderation is not included in the current version.
- **Push notifications**: Real-time browser push notifications or SMS alerts are not implemented; only email-based transactional notifications are supported.
- **Multi-language support**: The current implementation is in English only. Regionalisation and multi-language support are planned for future releases.

1.5 ORGANISATION OF THE REPORT

This report is organized into multiple chapters for systematic presentation of the proposed Smart Collaboration and Career Guidance System (CollabSphere). Chapter 1 introduces the project including background, problem statement, objectives, and scope. Chapter 2 presents the literature review and comparison of existing approaches. Chapter 3 discusses system analysis including existing system, proposed system, feasibility study, and requirements. Chapter 4 explains software design including architecture, UML diagrams, database design, and methodology. Chapter 5 describes implementation details and technologies used. Chapter 6 presents testing methodologies and results. Chapter 7 illustrates output screens and performance results. Finally, Chapters 8 and 9 provide conclusion and future scope respectively.

CHAPTER 2
LITERATURE REVIEW

2.1 OVERVIEW OF EXISTING WORK

Artificial Intelligence (AI) and digital platforms have significantly influenced modern educational technologies by enabling specialized career guidance and collaborative support systems. Several studies have explored web-based portals that analyse students' educational history, personal preferences, and academic records to recommend possible career options [1], [2]. Furthermore, machine learning algorithms have been deployed to develop intelligent recommendation systems that analyse user profiles, technical skills, and interests to offer personalized job role recommendations with improved accuracy [3]. Natural Language Processing (NLP) techniques have also been applied to develop intelligent chatbot systems that interpret students' queries in natural language, generating meaningful, context-aware career guidance and simulating conversations with an advisor [7]. In order to improve the effectiveness of career guidance, the integration of AI-based recommendation systems capable of analyzing huge amounts of user data has been found highly useful in identifying the best decision-making opportunities [6].

Alongside career recommendation engines, numerous platforms have been created to facilitate interaction and networking, offering mentorship opportunities where students learn from industry leaders [4]. To encourage skill development, various online learning platforms recommend customized courses and training based on user interests, allowing learners to acquire in-demand technical skills [5]. Researchers have also heavily investigated the role of collaborative learning environments—such as virtual study rooms and peer forums—in improving student academic outcomes, conceptual understanding, and motivation through peer-to-peer knowledge sharing [8]. Recent studies have further highlighted the importance of integrating real-time industry data, internship listings, and professional networking features directly into academic guidance platforms to bridge the gap between academic learning and industry expectations [9].

Despite these advancements, many existing systems operate in silos, focusing primarily on a single aspect of student development. Traditional recommendation portals often lack interaction features like communicating with mentors or collaborating with peers [2]. Machine learning models provide accurate job suggestions but usually omit real-time industry insights or collaborative knowledge-sharing platforms [3]. Mentorship platforms rely heavily on

manual interactions without offering automated data-driven recommendations [4], and online course platforms focus primarily on learning without integrated career planning [5]. Even NLP-powered career bots, while improving accessibility, typically lack integration with live mentors, collaborative study features, or real-time skill tracking [7]. Similarly, most collaborative study environments are entirely disconnected from career guidance systems [8], and direct industry integration remains largely absent in existing academic tools [9]. Consequently, a unified platform that comprehensively integrates AI career recommendations, professional mentorship, skill evaluation, and collaborative study features remains a critical missing component in the current educational technology landscape [6].

2.2 COMPARISON OF PREVIOUS APPROACHES

Table 2.1: Comparison of Existing Educational and Career Guidance Systems

Study / Tool Approach	Methodology	Limitation	Ref
Rule-Based Career Portals	Analyse academic records and educational history to recommend careers.	Recommends careers only; lacks collaborative environments, interaction features, or mentorship.	[1], [2]
Machine Learning Recommenders	Uses ML to analyse user profiles, technical skills, and interests for personalized recommendations.	Usually does not contain real-time industry insights or shared collaborative knowledge platforms.	[3]
Industry Mentorship Platforms	Facilitates manual networking and interaction with an industry panel.	Focuses on manual interactions; does not offer automated career recommendations through data analysis.	[4]
Adaptive Course Platforms	Recommends training and courses based on user interests.	Focuses primarily on learning/training; lacks personalized career planning or integrated guidance.	[5]
Standard AI Guidance Platforms	Analyse huge amounts of user data via AI to give career recommendations.	Unified integration of recommendation, mentorship, and collaboration is still absent.	[6]

NLP Career Chatbots	Uses NLP to interpret queries and simulate conversations with an advisor.	Lacks integration with live mentors, collaborative study features, or real-time tracking.	[7]
Virtual Collaborative Environments	Utilizes virtual study rooms and peer forums to foster peer-to-peer knowledge sharing.	Not linked to career guidance or mentorship systems, leaving a gap in holistic professional support.	[8]
Industry-Linked Academic Platforms	Integrates real-time internship data and professional networking directly into guidance.	This integration remains largely absent in existing tools, which operate in isolation.	[9]

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Current methods for student career planning, peer collaboration, and mentorship discovery primarily rely on a disjointed ecosystem of standalone applications. Students seeking peer study groups typically use generic instant messaging platforms (such as WhatsApp or Telegram) combined with basic file-sharing services (like Google Drive). For task management, they may separately employ tools like Trello or Notion. While these tools provide basic communication and organizational support, they operate entirely independently of one another and are not specifically tailored to academic collaboration.

Most existing systems allocate educational networking and career guidance to completely separate domains. Students looking for career advice rely on generic web searches, static educational portals, or standard generative AI models that lack context regarding structured academic transitions (e.g., Indian stream selection, localized degree programs). As a result, the advice received often remains generic, unverified, and fails to dynamically adapt to the student's specific academic background or interests.

Furthermore, conventional academic tools lack integrated mechanisms for professional mentorship. When students wish to seek professional guidance, they must manually discover mentors on social networking sites (like LinkedIn), coordinate session timings externally, and process payments using third-party UPI or banking apps without any escrow security or session tracking. This fragmented approach lacks the structured oversight, transparency, and integrated workflow necessary for a sustainable academic guidance ecosystem.

Limitations of Existing System

The major limitations of current educational collaboration and guidance systems are:

- Static, isolated collaboration tools without integrated task management.
- No contextual, domain-specific AI career guidance.
- Absence of structured mentor discovery and connection mechanisms.
- Lack of integrated, secure payment gateways for booking mentorship sessions.
- No role-based dashboards distinguishing between student and mentor workflows.

- Fragmented user experience requiring context-switching across multiple apps.
- Limited historical tracking of mentor-student engagement and peer collaboration.
- Inability to dynamically retrieve relevant career knowledge (no RAG implementation).

3.2 PROPOSED SYSTEM

The proposed system, CollabSphere (Smart Collaboration and Career Guidance System), is a unified, AI-powered web application designed to integrate collaborative study environments, intelligent career advisory, and professional mentorship into a single platform. Unlike fragmented traditional tools, the system leverages an integrated architectural approach to enhance academic efficiency, career clarity, and mentor accessibility.

The system collects user details upon encrypted, role-based registration, including name, bio, academic interests, and professional charges (for mentors). Based on this data, it sets up dedicated role-specific interfaces—a Student Dashboard for exploring courses, interacting with the AI, and finding mentors; and a Mentor Dashboard for tracking active students, earnings, and connection requests.

To facilitate collaboration, the system dynamically allows users to create subject-specific study rooms featuring real-time group chat and integrated Kanban boards (Todo / In Progress / Done) for shared task tracking. The system incorporates an advanced Artificial Intelligence layer—a Retrieval-Augmented Generation (RAG) pipeline utilizing the Groq Llama 3.1 LLM—which retrieves contextual career insights from a dedicated knowledge base to provide personalized, multi-turn career guidance. Furthermore, the system resolves the mentorship gap by integrating the Razorpay payment gateway (including Razorpay Route), enabling students to securely view mentor charges, book sessions per minute, and ensure funds are securely transferred via an escrow mechanism.

This integrated approach transforms isolated communication and static career searching into an adaptive, structured, and financially secure academic ecosystem.

Advantages of Proposed System

- Unified subject-based study rooms with integrated chat and Kanban tasks.
- Personalized, context-aware RAG AI chatbot for accurate career guidance.
- Secure mentor discovery and connection request tracking.

- Integrated Razorpay gateway for seamless per-minute session booking.
- Automated mentor payouts using Razorpay Route and escrow logic.
- Role-specific interactive dashboards preventing information overload.
- Real-time room membership management and access control.
- Centralized profile and interest tracking for both students and mentors.
- Modular and scalable modern web-based implementation.

3.3 FEASIBILITY STUDY

The feasibility study evaluates the practicality and viability of implementing the proposed CollabSphere system.

Technical Feasibility

The proposed system is technically feasible as it is developed using widely adopted modern web technologies such as React 19, Vite, Node.js (for frontend tools), Python, Flask, and PostgreSQL. These technologies provide sufficient flexibility and robust support for implementing secure REST APIs, real-time database querying, and JWT-based authentication. The integration of advanced features relies on established standard APIs (Groq API for LLM completion and Razorpay SDK for financial transactions), meaning the required cloud integrations are readily compatible with the proposed stack without necessitating extreme local computational horsepower.

Economic Feasibility

The project is economically feasible because it does not require purchasing expensive hardware or proprietary enterprise software licenses. The development frameworks (React, Flask, PostgreSQL) are open-source and freely available. Deployment and backend API costs are minimal due to the availability of generous developer free-tiers (like Vercel and Groq). Additionally, the Razorpay payment architecture operates on a commission-per-transaction model, avoiding heavy upfront licensing costs, making the platform financially sustainable.

Operational Feasibility

The proposed system is operationally feasible because it is designed with a clean, glassmorphic, and highly intuitive user interface that aligns with modern web standards, meaning it can be easily operated by students and mentors without requiring technical expertise. By centralizing

features that users previously had to find across five different apps, the workflow aligns perfectly with the natural academic habits of users, almost ensuring practical real-world adoption.

3.4 REQUIREMENTS

3.4.1 Hardware Requirements

The CollabSphere platform is a cloud-hosted web application, meaning end-users do not require specialized hardware. The minimum hardware requirements for development and deployment are as follows:

- Processor: Intel Core i3 / AMD Ryzen 3 or higher
- RAM: Minimum 4 GB (8 GB recommended for development environment)
- Storage: Minimum 10 GB of available disk space
- Internet Connection: Stable broadband or 4G connection
- Display: Standard monitor with minimum resolution of 1366 × 768

For development and compilation purposes, particularly for the React frontend, a system with 8 GB RAM and SSD storage is recommended to ensure smooth execution of local servers.

3.4.2 Software Requirements

The system is developed using a modern dual-stack architecture (JavaScript frontend and Python backend). The required software components include:

- Operating System: Windows 10/11 / Linux (Ubuntu) / macOS
- Frontend Library: React 19
- Backend Framework: Python Flask
- Database: PostgreSQL (Production) / SQLite (Development)
- Runtime Environment: Node.js (for frontend build tools)
- Package Managers: npm (Node) / pip (Python)
- Build Tool: Vite
- Code Editor: Visual Studio Code

- Web Browser: Google Chrome / Mozilla Firefox / Safari
- Version Control: Git

3.4.3 Functional Requirements

The system shall:

- Allow users to register and authenticate via JWT with designated roles (Student or Mentor).
- Permit users to reset forgotten passwords securely via email recovery links.
- Enable students to create subject-specific study rooms and manage membership join requests.
- Allow room members to create, edit, assign, and track tasks across Todo/In Progress/Done columns.
- Provide persistent group chat functionalities for all members within a specific study room.
- Allow students to converse with an AI career guidance chatbot powered by a RAG pipeline.
- Enable students to view a directory of available mentors, including bios and per-minute charges.
- Allow students to send customized connection requests to mentors, which mentors can approve or reject.
- Permit students to initialize secure Razorpay payment orders to book mentorship sessions.
- Provide mentors with automated payout onboarding (Razorpay Route) and maintain full transaction history.
- Give users access to role-specific dashboard navigation components.

3.4.4 Non-Functional Requirements

- Performance: The system shall respond to database queries and API calls rapidly, ensuring a smooth user experience even during complex chatbot inferences.

- Usability: The interface shall feature responsive, modern glassmorphic design principles that render correctly on desktop and tablet screens.
- Security: The system shall securely hash all user passwords utilizing bcrypt algorithms and encrypt all payment transmissions using Razorpay logic. JWT tokens must be utilized to protect all private API routes.
- Reliability: The backend and database connections shall handle multiple concurrent room operations and chat posts without localized crashes or data corruption.
- Scalability: The Flask Blueprint routing architecture and React component tree shall be structured to allow for seamless future feature additions.
- Maintainability: The codebase shall adhere to modular design principles with strict separation between routes, controllers, and models.
- Compatibility: The web application shall function properly across all modern HTML5-compliant web browsers without layout breakage.

CHAPTER 4

SYSTEM DESIGN

4.1 ARCHITECTURE DESIGN

The architecture of the proposed Smart Collaboration and Career Guidance System (CollabSphere) is designed using a modular, layered structure (client-server architecture) to ensure scalability, maintainability, and seamless performance across multiple user roles. The system is divided into Several interconnected components including the Frontend Dashboard Interface, User Authentication & Role Management Module, Collaborative Study Room Module, Task Management Module, AI Career Guidance Module, and the Mentor Connection & Payment Module.

The Frontend Dashboard Interface serves as the primary communication layer, presenting role-specific analytics, schedules, and tools to Students and Mentors in an interactive, glassmorphic layout. The User Authentication & Role Management Module securely handles registration, JWT-based logic, email recovery, and profile configurations. The Collaborative Study Room Module enables the creation of subject-specific peer environments and manages real-time group chat state and access controls. The Task Management Module operates within these rooms to process Kanban-style assignments, allowing users to track academic progress seamlessly.

Driven by advanced data processes, the AI Career Guidance Module utilizes a Retrieval-Augmented Generation (RAG) pipeline to fetch contextual passages from a structured career knowledge base, passing them to the Groq LLM to provide highly personalized, multi-turn career advice. Finally, the Mentor Connection & Payment Module processes secure interactions between mentees and professionals, managing connection approvals, tracking mentorship sessions, and processing financial transactions securely through Razorpay's escrow and webhook infrastructure.

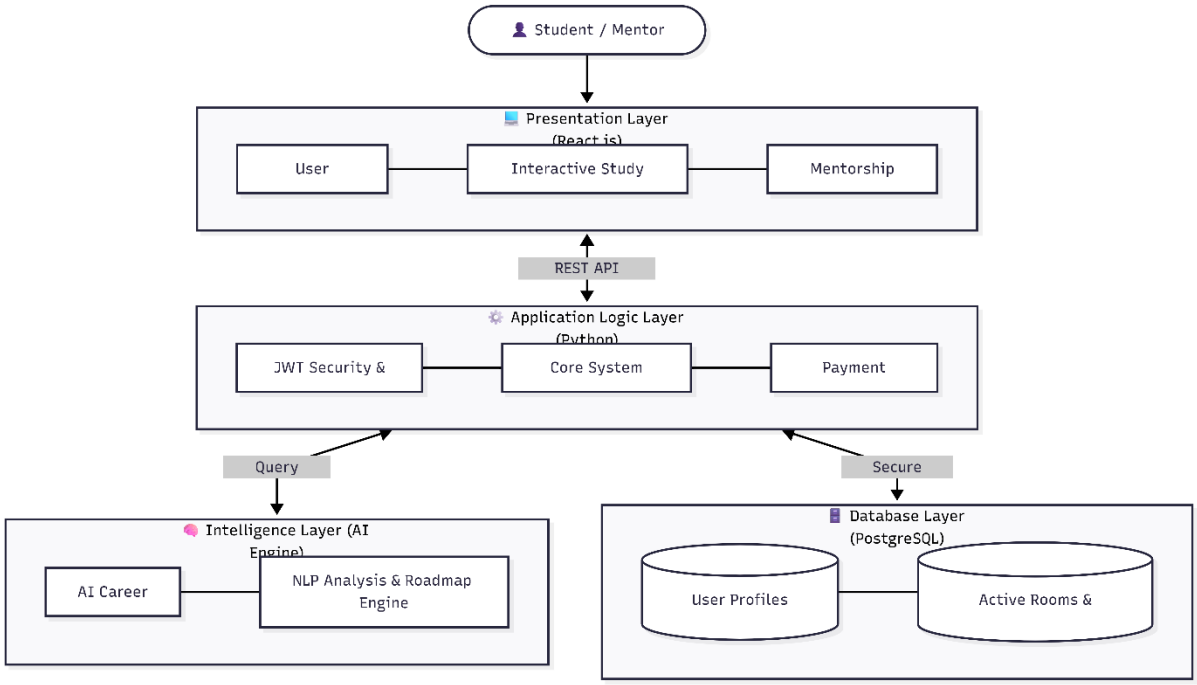

Fig 4.1: Architecture of the Smart Collaboration and Career Guidance System (CollabSphere)

4.2 UML Diagrams

4.2.1 Class Diagram

Objective:

Define the static structure of the AI-Based Career Guidance and Collaborative Learning Platform, including classes, their attributes, methods, and the relationships between system components.

Description:

Classes: User, StudentProfile, AIRecommendationEngine, CareerPathway, StudyRoom, MentorshipSession, CareerBotService, DashboardAnalytics

User – Represents an individual registered on the platform.

Attributes: id, name, email, passwordHash, role (student/mentor), createdAt

Methods: register(), login(), updateDetails()

StudentProfile – Stores the academic and skill-based data of a student.

Attributes: userId, education, technicalSkills, interests, assessmentScores

Methods: updateProfile(), submitAssessment(), getStrengths()

AIRecommendationEngine – Core module responsible for analyzing profiles and predicting ideal careers.

Attributes: profilingData, marketTrendsData

Methods: analyseProfile(), predictCareerRoles(), generateActionablePathway()

CareerPathway – Represents the generated roadmap for a specific career role.

Attributes: careerName, recommendedCourses, certifications, projectIdeas

Methods: fetchResources(), trackPathwayProgress()

StudyRoom – Handles collaborative pear-to-peer learning sessions.

Attributes: roomId, topic, activeMembers, chatHistory

Methods: createRoom(), joinRoom(), sendMessage()

MentorshipSession – Manages the networking between students and industry mentors.

Attributes: sessionId, studentId, mentorId, scheduledDate, status

Methods: requestSession(), approveSession(), completeSession()

CareerBotService – Provides conversational AI counseling and guidance.

Attributes: apiKey, contextMemory

Methods: processQuery(), generateGuidanceResponse()

DashboardAnalytics – Processes progress data to generate visual insights.

Attributes: userProgress, courseCompletionRate, skillGrowth

Methods: generateProgressReport(), visualizeGrowth()

Relationships:

User owns a StudentProfile.

AIRecommendationEngine uses StudentProfile data to generate CareerPathways.

StudyRoom and MentorshipSession facilitate collaborative interactions for the User.

CareerBotService assists the User with real-time NLP guidance.

DashboardAnalytics aggregates data from CareerPathway and User interactions.

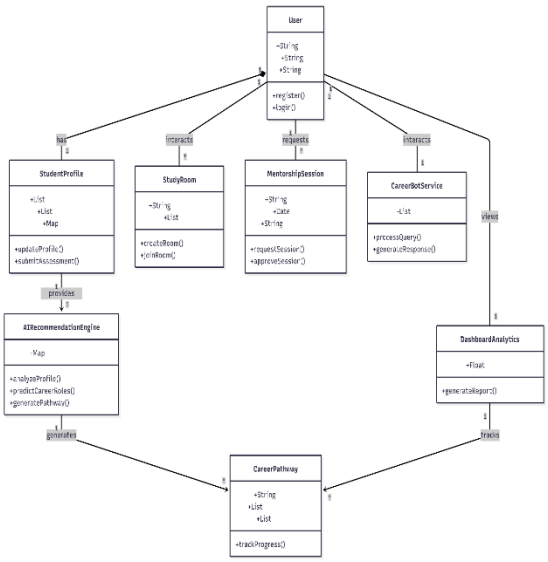

Fig 4.2.1. Class Diagram

4.2.2 Interaction Diagrams

4.2.2.1 Sequence Diagram

Objective: Show how objects interact in a particular sequence to generate career recommendations and collaborative opportunities.

Actors: Student, Dashboard Interface, Profile Manager, AI Recommendation Engine, Collaborative Module, CareerBot Service.

Sequence:

Student registers, completes their profile, and takes the technical and problem-solving assessment.

Dashboard Interface sends the profile and assessment scores to the Profile Manager.

Profile Manager securely stores the student's data.

AI Recommendation Engine retrieves the profile data and evaluates technical skills against market demands.

AI Recommendation Engine generates targeted career roles (e.g., Data Scientist, Software Developer) and actionable learning pathways.

Collaborative Module matches the student with relevant virtual Study Rooms and available Industry Mentors based on their chosen career path.

CareerBot Service initializes to provide real-time Q&A support regarding the generated pathway.

Dashboard Interface displays the AI-generated career suggestions, learning courses, and collaborative networking options to the Student.

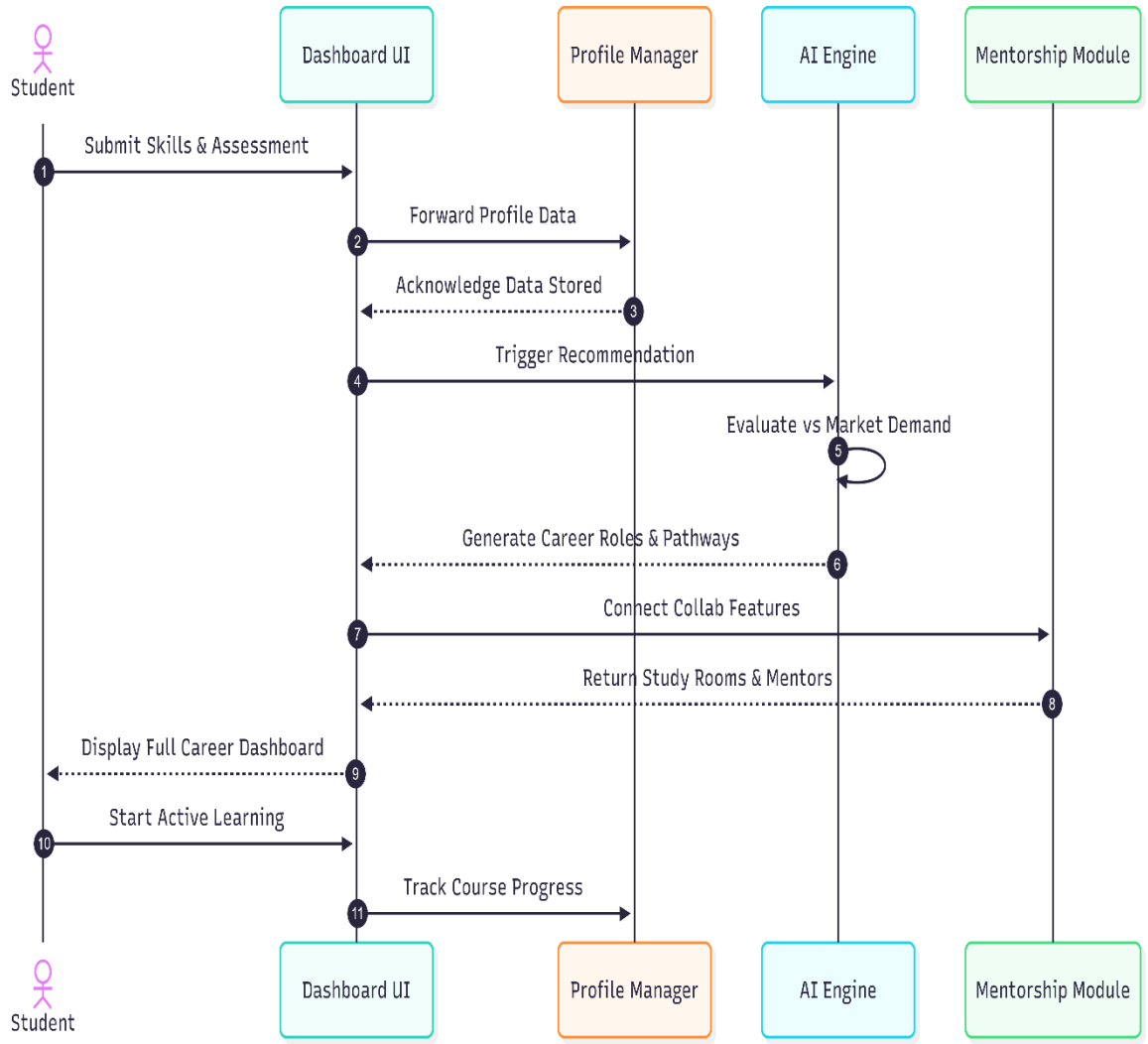

Fig. 4.2.2.1 Sequence Diagram

4.2.2.2 Collaboration Diagram

Objective: Show how system components collaborate with each other to complete the task of profiling a student and providing career pathways.

Description:

Actors: Student, Dashboard Interface, Profile Manager, AI Recommendation Engine, Mentorship Module, Analytics Module.

Interactions:

Student submits educational background, skills, and assessment answers.

Dashboard Interface forwards this comprehensive data to the Profile Manager.

Profile Manager stores the data and triggers the evaluation phase.

AI Recommendation Engine retrieves the data, analyse strengths, and predicts the best-fit career roles.

AI Recommendation Engine generates a tailored roadmap including courses and certifications.

Mentorship Module suggests industry mentors aligned with the recommended careers.

Analytics Module tracks the student's interaction with the suggested courses to update long-term progress.

Dashboard Interface presents the unified collaborative and career guidance view to the Student.

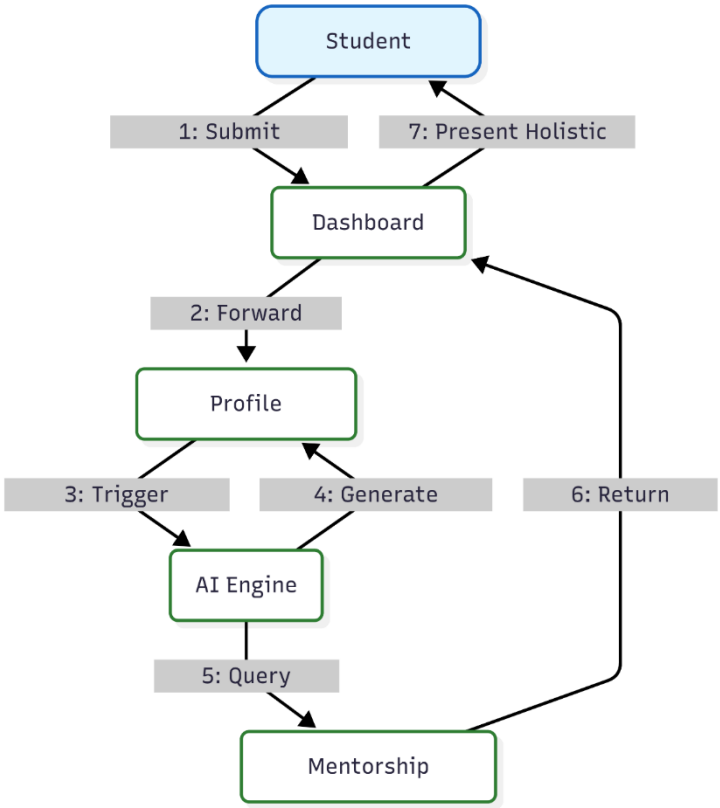

Fig. 4.2.2.2 Collaboration Diagram

4.2.3 Object Diagram

Objective: Represent a snapshot of objects in the AI-Based Career Guidance and Collaborative Learning Platform at a particular moment and illustrate how instances of classes interact with each other.

Objects: StudentUser, StudentProfile, AI_Engine, TechPathway, SoftwareDevMentor, ProgressAnalytics.

Interactions:

StudentUser object creates a StudentProfile instance by submitting skills and test results.

AI_Engine object reads the StudentProfile instance to evaluate technical proficiencies.

TechPathway object is instantiated by the AI, containing a roadmap for "Software Developer" with specific React and Python courses.

SoftwareDevMentor object receives a connection request triggered by the user exploring the TechPathway.

ProgressAnalytics object monitors the StudentUser's completion of courses within the pathway to generate UI charts.

StudentUser object views all suggestions, mentor connections, and progress charts on the dashboard.

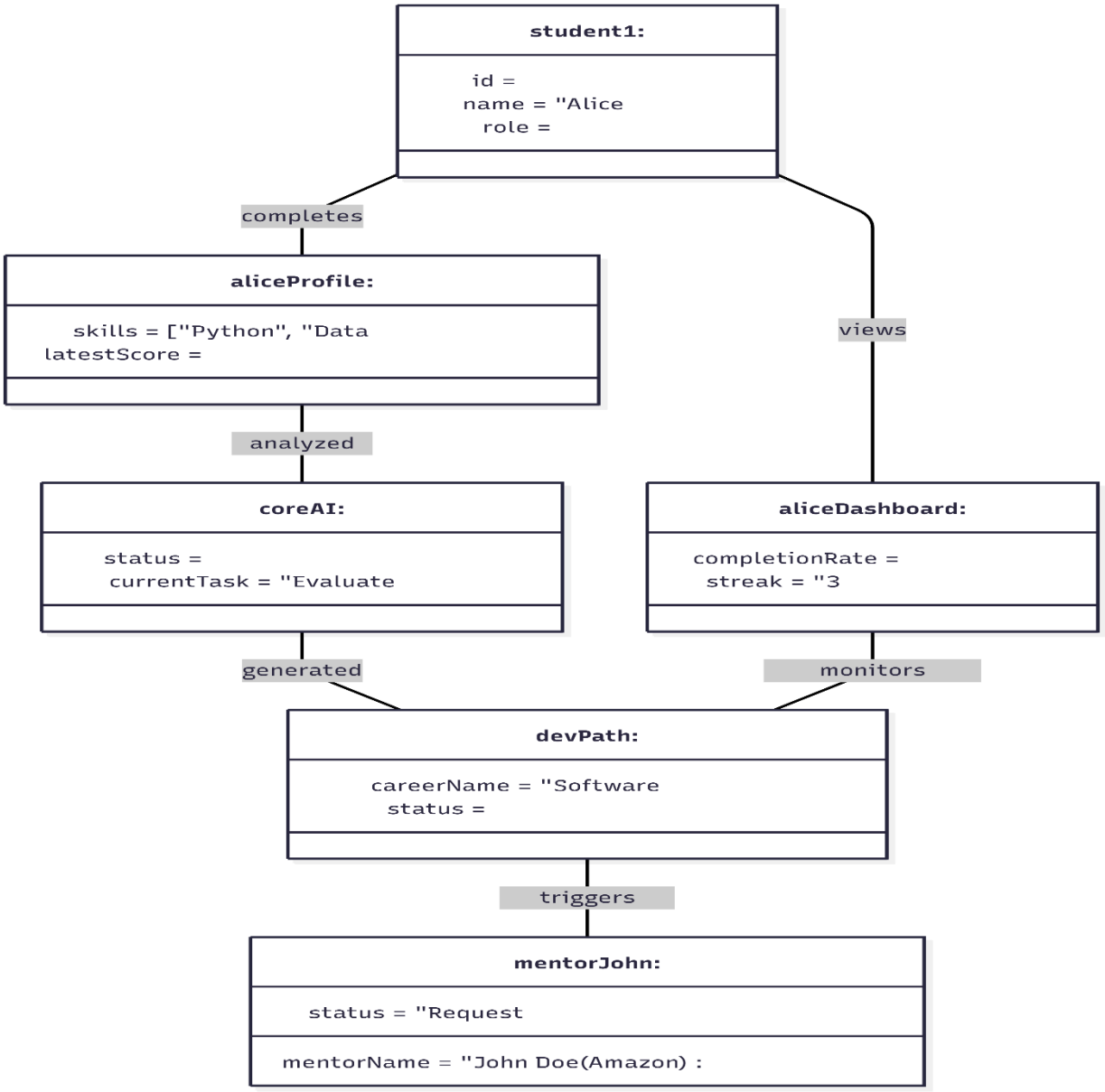

Fig. 4.2.3 Object Diagram

4.2.4 Use Case Diagram

Objective: Illustrate the functional interactions between the user and the system by representing the key system functionalities.

Actor: Student, Industry Mentor

Use Cases:

Register and Build Profile – The student enters personal information, skills, and interests.

Take Skill Assessment – The student completes technical and problem-solving tests.

Generate Career Recommendations – The AI system processes data to suggest roles like Cyber Security or Cloud Computing.

Explore Learning Pathways – The student views curated courses, certifications, and project ideas.

Join Virtual Study Rooms – The student collaborates with peers in discussion forums and group projects.

Request Mentor Guidance – The student sends requests to industry mentors for real-world professional advice (Mentors can Accept/Reject).

Interact with Career Bot – The student chats with the AI for dynamic, localized guidance.

Track Career Progress – The system tracks skill improvements over the years and redirects guidance dynamically.

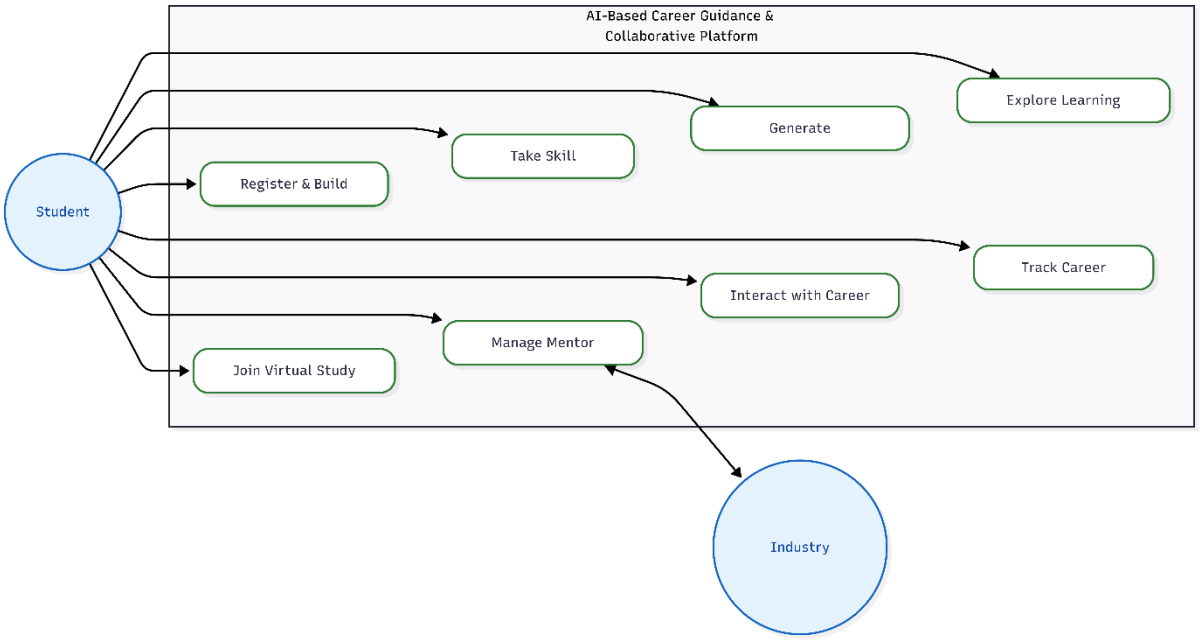

Fig. 4.2.4 Use Case Diagram

4.2.5 Activity Diagram

Objective: Represent the workflow of the platform by illustrating the sequence of activities from profile creation to collaborative learning and progress tracking.

Activities: User Input, Skill Profiling, AI Analysis, Pathway Generation, Collaborative Networking, Progress Tracking.

Flow:

Student logs into the application and enters education details, technical skills, and takes an assessment.

The Profile Manager securely stores the data and creates a comprehensive student profile.

The AI Recommendation Engine analyses the profile to identify the candidate's strengths and weaknesses.

The AI Recommendation Engine outputs suitable career options (e.g., Data Scientist) alongside specific learning pathways (courses, tutorials).

The system simultaneously queries the database to suggest relevant Discussion Forums, Virtual Study Rooms, and Industry Mentors.

The Student engages with the courses and connects with peers and mentors.

The Analytics Module processes the student's ongoing activities and courses completed to update their growth chart.

The system dynamically recalculates the student's profile and tailors future guidance based on newfound skills, repeating the feedback loop.

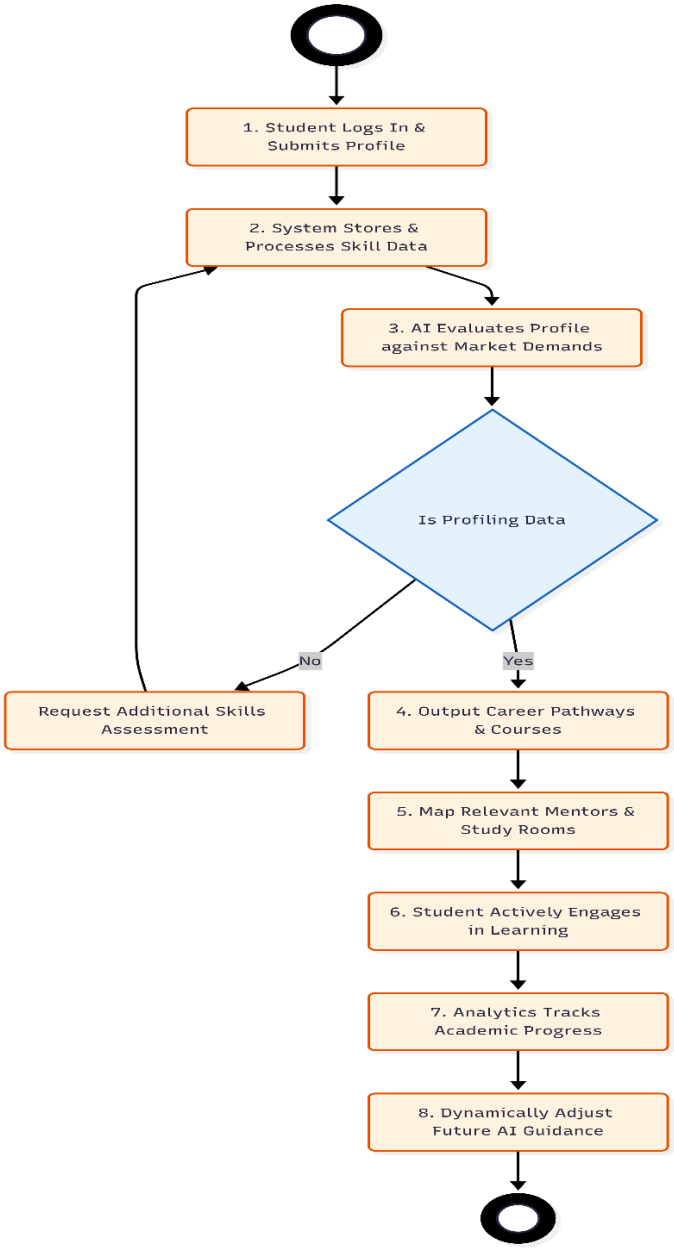

Fig. 4.2.5 Activity Diagram

4.2.6 Control Flow Diagram

Objective: Represent the flow of control within the system from initial student assessment to complete career guidance and result visualization. Description:

Components: Student, Profile Manager, AI Recommendation Engine, Collaborative Module, Dashboard Interface, Analytics Tracker.

Flow:

Student inputs personal details, technical knowledge, and completes assessments through the Dashboard Interface.

Profile Manager processes and structures the data for AI ingestion.

AI Recommendation Engine evaluates problem-solving logic, skills, and market trends to generate accurate career matches.

AI Recommendation Engine dynamically constructs customized learning pathways containing courses and projects.

Collaborative Module intercepts the career choice to load relevant study rooms, forums, and mentor profiles.

Analytics Tracker initializes a baseline for the student's current skill level.

Dashboard Interface displays the holistic view: AI career choices, courses, peer rooms, and mentor connections.

Student actively learns and networks, updating their completion status, which triggers the Analytics Tracker to update progress graphs and dynamically refine future AI recommendations.

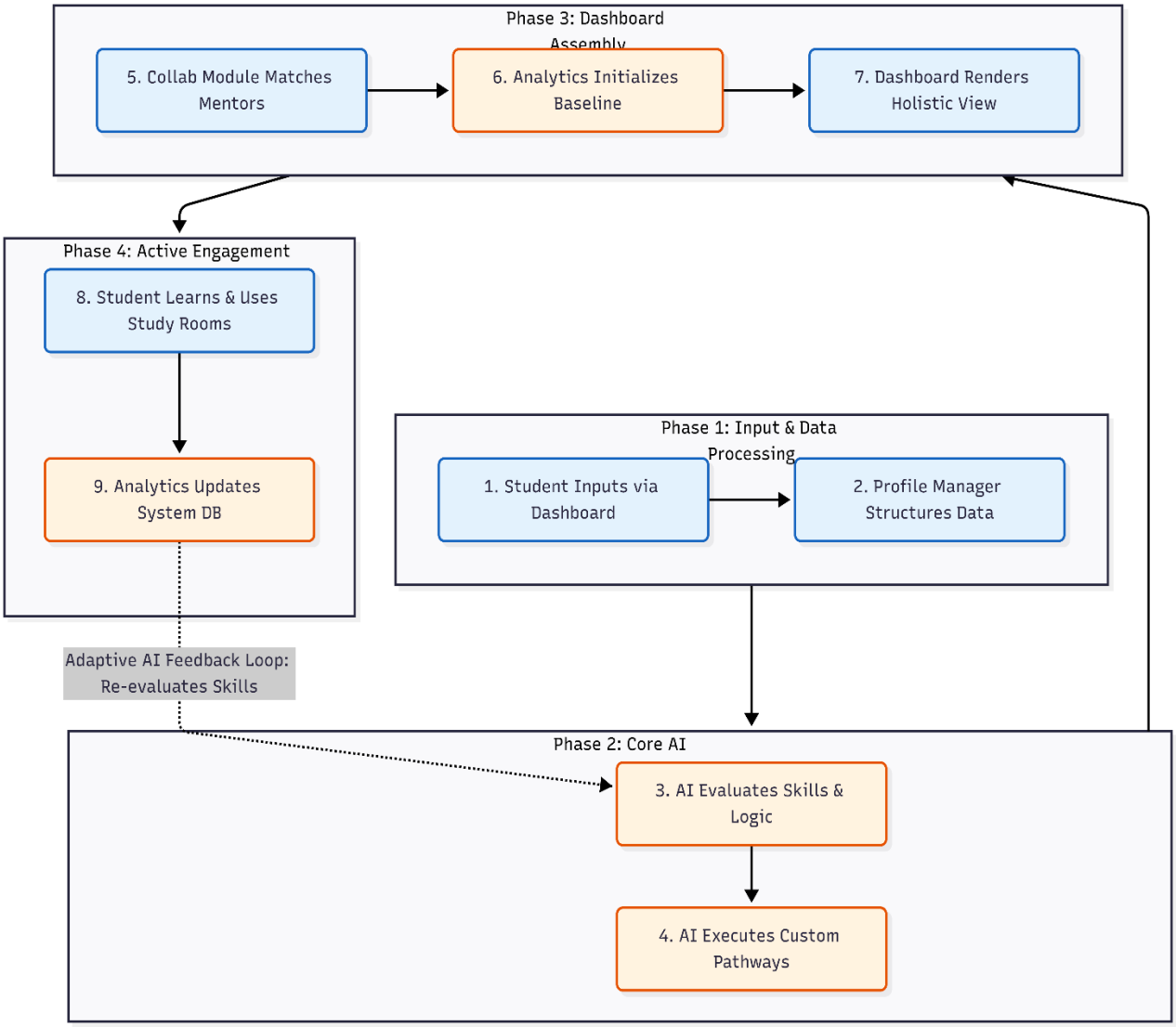

Fig. 4.2.6 Control Flow Diagram

4.3 DATABASE DESIGN

The proposed Smart Collaboration and Career Guidance System (CollabSphere) utilizes a robust relational database architecture managed via SQLAlchemy Object Relational Mapper (ORM), with PostgreSQL targeted for production and SQLite for local development. This approach guarantees data integrity, secure transactional operations, and strict relational continuity between users, collaborative workspaces, and financial records.

The primary data entities stored within the system include Users (Students and Mentors), Rooms, Room_Members, Tasks, Messages, Mentor Connections, and Payments.

Table 4.1: User Schema

Field Name	Data Type	Description
id	Integer (Primary Key)	Unique identifier for user
email	String	Unique email address for login
password	String	Bcrypt hashed password
name	String	Full name of the user
role	String	Role designation ('student' or 'mentor')
charge_per_min	Float	Rate billed per session (Mentors only)
razorpay_account_id	String	Associated Route payout ID (Mentors only)

Table 4.2: Room Schema

Field Name	Data Type	Description
id	Integer (Primary Key)	Unique identifier for study room
subject	String	Primary topic of the study room
description	String	Detailed purview of the collaborative space
creator_id	Integer (Foreign Key)	Reference to the User ID who initiated the room
created_at	DateTime	Timestamp of room creation

Table 4.3: Payment Schema

Field Name	Data Type	Description
id	Integer (Primary Key)	Unique transaction identifier
student_id	Integer (Foreign Key)	User ID identifying the payer
mentor_id	Integer (Foreign Key)	User ID identifying the payee
amount_paise	Integer	Total transaction volume (₹1 = 100 paise)
razorpay_order_id	String	Verified Order token from Razorpay Gateway
status	String	Payment status ('created', 'paid', 'failed')

4.4 ALGORITHM / DESIGN METHODOLOGY

The proposed system involves multiple complex pipelines to handle different interactions. The primary algorithm governing the system's most complex feature—the AI Career Guidance Bot—is the **Retrieval-Augmented Generation (RAG) Algorithm**. This dynamic methodology generates context-aware career recommendations by blending localized structured data with large language models.

RAG Pipeline Methodology:

1. **Document Ingestion:** The backend system loads a localized text-based knowledge base (career_docs.txt) analyzing Indian stream selections, courses, and industry trends, breaking it into smaller searchable paragraphs or "chunks".
2. **Query Tokenization:** When a student enters a career query, the system strips standard "stop words" and tokenizes the request into core keyword elements.
3. **TF-IDF Semantic Scoring:** The core AI service calculates a Term Frequency-Inverse Document Frequency (TF-IDF) score for all chunks in the text document compared to the query tokens, isolating the exact paragraphs that contain the most relevant career information.
4. **Context Injection:** The top three highest-scoring contextual passages are injected directly into a structured System Prompt.
5. **LLM Generation:** The enriched prompt alongside the last 6 responses of the user's conversational history is transmitted securely to the Groq Llama 3.1 LLM via secure API calls.
6. **Delivery:** The refined, contextualized guidance is returned and streamed via the database directly into the student's intuitive chatbot interface.

This methodology ensures that the AI guidance mechanism operates tightly within grounded, reliable academic boundaries rather than hallucinating generic responses, maximizing student reliability and system trust.

CHAPTER 5

IMPLEMENATION &CODING

5.1 TECHNOLOGIES USED

Frontend: React 19, Vite, React Router, Axios, Firebase, Tailwind CSS, Lucide icons

Backend: Flask, SQLAlchemy, JWT, PostgreSQL, Razorpay, Groq (Llama 3.1 AI)

5.2 MODULES DESCRIPTION

The proposed system is organized into two main modular parts: Backend and Frontend. The backend is responsible for handling AI-related services, relational database management, and API communication, while the frontend manages the user interface, navigation, skill assessments, collaborative networking features, and overall interaction with the system. This modular stricture helps in better development, maintenance, and future enhancement of the application.

A. Backend

The backend module contains the server-side logic of the system. It is primarily responsible for processing AI-related inferences, securely storing student profiles in the database, evaluating mentor requests, and connecting the frontend with language model services. *Main* **Files included in Backend:**

- `app.py`
- `models.py`
- `extensions.py`
- `routes/auth_routes.py`
- `routes/chat_routes.py`
- `routes/user_routes.py`
- `routes/connection_routes.py`
- `services/ai_bot.py`

B. Frontend

The frontend module contains the client-side architecture built on React.js. It is responsible for displaying the career guidance interface, managing dynamic tab-based navigation, securely storing session tokens, and presenting AI-generated pathways to the user. *Main files* **included in Frontend:**

- `src/main.jsx`
- `src/App.jsx`
- `src/contexts/AuthContext.jsx`

Frontend Pages:

- `src/pages/Login.jsx`
- `src/pages/StudentDashboard.jsx`
- `src/pages/MentorDashboard.jsx`
- `src/pages/RoomBoard.jsx`

Frontend Components:

- `src/components/student/DashboardTab.jsx`
- `src/components/student/ProfileTab.jsx`
- `src/components/student/CareerBotTab.jsx`
- `src/components/student/FindMentorTab.jsx`
- `src/components/student/CoursesTab.jsx`
- `src/components/student/RoomDetailView.jsx`

C. Styling

The styling section contains the design-related files used to maintain the visual appearance, glassmorphic UI, animations, and responsiveness of the application. *Style file:*

- `src/index.css`

5.2 Backend

File: routes/chat_routes.py

Purpose:

Handles all backend API routes for the generative AI features of the platform. It accepts requests from the React frontend, formats the prompts, sends data to the AI service layer (ai_bot.py), and returns the intelligent guidance.

Main Functions:

- Generates responses to career-related student inquiries.
- Explains specific technical skills required for jobs.
- Suggests actionable learning resources dynamically.
- Maintains secure chat history. **Code Summary:** The file imports Flask Blueprints and AI extraction methods. All routes use the POST method since conversation context is sent dynamically in the request body.

Routes Included:

- /api/chat/ask – Accepts a student's prompt and returns the AI's explanation. **Error Handling:** Each route uses try-catch blocks to prevent crashes. If the external language model times out or returns an invalid structure, a fallback error message is safely returned to the user interface.

File: services/ai_bot.py

Purpose:

Handles all core AI-related backend logic of the platform. It communicates securely with the external Language Model, constructs localized prompts mapping to a student's profile, and formats strings for database ingestion.

Overview:

This file acts as the AI Service Layer, keeping complex prompt-engineering logic separate from standard HTTP routing files for much better organization and maintainability. **Main Components:**

- analyse_profile(data) – Evaluates student test scores and maps competencies.
- predict_career_roles() – Uses assessment data to predict career matches like Data Scientist or Cloud Architect.

- generate_pathway(data) – Curates an actionable list of skills, certifications, and projects tailored to a specific career recommendation.
- process_conversation() – The core conversational engine maintaining context for the CareerBot.

Special Feature:

The pathway generation function ensures robust formatting. It specifically forces the AI to output verified JSON arrays, strips away random markdown characters, and safely parses the block so the frontend React components don't crash when rendering the roadmap.

Error Handling:

Includes strict API validation, status code checking, and fallback except Exception as e handling to log internal errors securely.

File: app.py

Purpose:

Acts as the main operational entry point of the Python backend application. It initializes the Flask web server, hooks into the SQL database, loads environment settings, and binds the routing blueprints.

Overview:

This file prepares the backend so the frontend can retrieve data. It is the very first file executed upon system startup.

Main Functions:

- Loads sensitive environment variables (like API Keys and DB credentials) using dotenv.
- Creates the central Flask application instance.
- Enables Cross-Origin Resource Sharing (CORS) so the Vite frontend can communicate securely.
- Registers route blueprints (/api/auth, /api/chat, /api/rooms).

- Configures and starts the backend server asynchronously. **Role in the Project:** It serves as the essential backbone of the backend system by handling server setup, API connection points, and communication between React and Python.

File: models.py

Purpose:

Defines the relational schema of the entire platform using Object-Relational Mapping (ORM) techniques. It dictates exactly how data is stored, related, and fetched from the PostgreSQL database.

Overview:

Maintains data integrity by defining required fields, foreign key constraints (like linking a Mentor Request to a User ID), and validation parameters for profiles and rooms.

Role in the Project:

Ensures all student academic records, collaborative chat histories, and mentor connections persist safely long-term.

Frontend

File: ProfileTab.jsx

Purpose:

Implements the profiling module of the guidance platform where students manage their educational background, technical skills, and assessment scores.

Overview:

This component loads securely stored database information upon login. It enables the foundational data collection step needed by the AI recommendation engine.

Main Functions:

- Loads existing academic data using useEffect.
- Stores edits locally via React useState.
- Includes a secure submission function making a PUT request to /api/users/profile.

UI Elements:

- Responsive form grid for personal information.
- Tag-based input for technical skills (e.g., Python, C++).
- Test score visualizers.
- Save button with loading-state spinners.

Role in the Project:

It feeds the analytical core. Without the robust data recorded here, the AI cannot generate accurate career pathways.

File: DashboardTab.jsx

Purpose:

Implements the primary dashboard of the platform, providing an immediate visual overview of study progress, generated career recommendations, and active mentorships.

Overview:

This component acts as the homepage post-login. It loads a student's profile analytics, structures their AI-generated career paths, and provides quick navigation options to collaboration hubs.

Main Functions:

- Loads the user's AuthContext to pull profile names and statistics.
- Dynamically maps over the array of AI-recommended target careers.
- Calculates summary metrics like active courses and pending mentor requests.

UI Elements:

- A personalized welcome banner.
- Summary statistic cards with engaging icons.
- Action-oriented buttons urging the student to "Find Mentor" or "View Courses".

Role in the Project:

It serves as the central control panel, summarizing status, highlighting actionable pathways, and improving seamless navigation across the collaborative ecosystem.

File: FindMentorTab.jsx

Purpose:

Implements the professional networking and peer connection aspect of the platform. It matches students with industry experts relative to their generated career field.

Overview:

Instead of isolating students behind tutorials, this feature lists highly relevant professionals.

Main Functions:

- Retrieves all verified mentors from the database.
- Allows sorting and filtering based on the student's chosen roadmap.
- Handles the logic to trigger a "Connection Request" API call to the backend.

UI Elements:

- Professional Mentor Cards detailing company, job title, and expertise.
- Interactive Connect buttons that toggle state dynamically upon sending a request.

Role in the Project:

It bridges the critical gap between academic learning and real-world employment networking.

File: RoomBoard.jsx

Purpose:

Houses the collaborative, synchronized learning environment of the platform.

Overview:

Rather than studying alone, this module renders virtual study rooms linked to specific technological frameworks or course topics.

Main Functions:

- Retrieves room history and live member connections.

- Processes instant messaging functionality and peer-to-peer file sharing interactions.
- Validates user permissions before allowing access.

UI Elements:

- Real-time chat window and text-input area.
- A sidebar displaying currently active group members and active discussions.

Role in the Project:

It represents the "Collaborative" pillar of the system—promoting teamwork and group projects.

File: CareerBotTab.jsx

Purpose:

Implements the Generative AI Explanation module on the frontend.

Overview:

This component provides a familiar conversational chat interface mimicking ChatGPT. It seamlessly links the student to the chat_routes.py backend.

Main Functions:

- Manages the local state array of user and bot messages.
- Automatically scrolls to the newest message.
- Disables the input field while waiting for the AI response to prevent duplicate API calls.

UI Elements:

- Scrollable chat history window.
- Distinct styling distinguishing user bubbles from AI system bubbles.
- A fixed input bar at the bottom with a submit button.

Role in the Project:

It enhances explainability, ensuring that a student always has a virtual counselor to ask "Why should I learn this?" or "Can you further explain this career?".

•

File: App.jsx

Purpose:

Defines the client-side routing hierarchy and ensures application security.

Overview:

This file acts as the central navigation logic wrapper of the platform. It uses react-router-dom to dictate how URLs map to specific React components.

Main Functions:

- Validates secure ProtectedRoute wrappers around internal dashboards.
- Renders shared persistent layouts like navigation sidebars and top-bars.
- Gracefully routes unauthorized users back to the Login.jsx screen.

Role in the Project:

It acts as the navigation control layer, managing secure page flow, routing hierarchy, and component lifecycles.

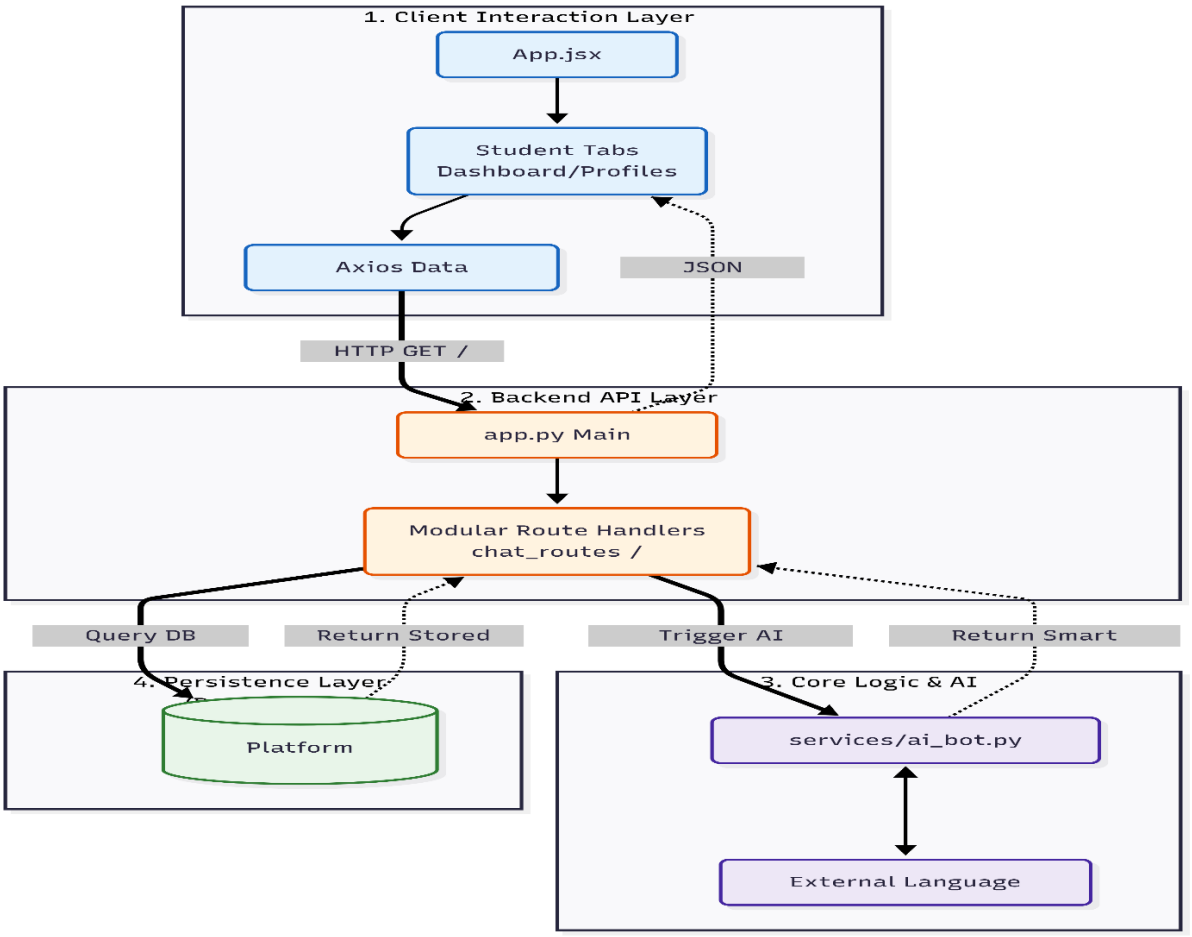

Fig 5.1. Layered Execution Flow of the AI-Based Career Guidance (This diagram represents the layered architectural flow of the system. It explains how React layout components guide the user to functional modules, how frontend tabs perform state management operations, and how Python backend routes and AI services collaborate to recommend pathways and return the final explainable output to the student.)

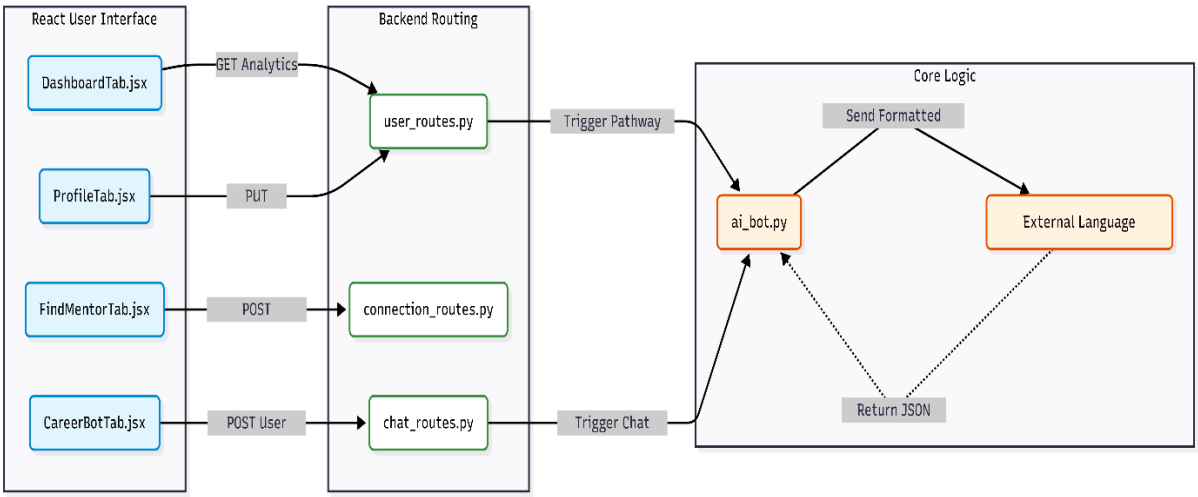

Fig. 5.2. File Interaction Flow of the AI-Based Career Guidance (This diagram illustrates the file-level interaction between React User Interface and the Backend .)

CHAPTER 6

TESTING

6.1 TEST CASES

Test cases were designed and executed to validate the major functionalities of the Smart Collaboration and Career Guidance System (CollabSphere). The purpose of these test cases was to ensure that the system performs securely and accurately across various user inputs, interactions, and operational scenarios involving both student and mentor workflows.

Table 6.1: Test Cases of Smart Collaboration and Career Guidance System

Test Case ID	Test Scenario	Input	Expected Output	Result
TC01	User Authentication	Valid email & password	Issues JWT Token and logs user into dashboard	Passed
TC02	Create Study Room	Room subject & description	Subject room created and visible under Rooms tab	Passed
TC03	Task Kanban Move	Drag task to 'In Progress'	Task status updates in database and reflects on UI	Passed
TC04	Mentor Connection	Select mentor and send note	Connection request status set to pending	Passed
TC05	Accept Mentorship	Mentor clicks 'Approve'	Status updates and payment option unlocks for student	Passed
TC06	Payment Integration	Credit Card details submitted	Razorpay processes transaction and verifies signature	Passed
TC07	RAG AI Chatbot	Ask about "Career post-10th"	Contextual advice returned using local career docs	Passed
TC08	Group Chat Message	Send text in study room	Message broadcasts and persists in database	Passed

6.2 TESTING METHODS

6.2.1 Unit Testing

Unit Testing is the process of testing individual components or functions independently to ensure that each isolated module performs correctly. Examples of units tested in CollabSphere include:

- Validation of login and registration input fields, including password hashing checking.
- Addition, moving, and deletion of Kanban board tasks.
- Tokenization and TF-IDF extraction logic inside the RAG AI script.
- Routing protections restricting student access from mentor-specific pages.
- Calculation logic determining per-minute payment totals based on study durations.

Unit testing vastly improved the reliability of individual modules and reduced the possibility of localized errors migrating into later stages of application development.

6.2.2 Integration Testing

Integration Testing is performed to verify whether different modules, database structures, and third-party APIs work correctly when combined together. The following integrations were tested:

- Communication between React frontend components and Flask REST APIs over Axios.
- Integration of stored task and chat data between PostgreSQL database models.
- Secure transmission of prompts and contextual knowledge chunks to the Groq Llama 3.1 LLM via encrypted API calls.
- End-to-End connection involving Razorpay checkout initiation, user payment, cryptographic signature verification, and automated webhook triggers.

This testing phase ensured that all internal components and external service providers were properly linked and delivered a consistent, crash-free user experience.

6.2.3 System Testing

System Testing is conducted to evaluate the complete and integrated software application as a whole from the perspective of an end-user. System testing verified that:

- The application correctly executes the full "Student to Mentor" lifecycle (Login → Request Mentor → Mentor Approval → Payment processing).
- The system behaves predictably under different roles, correctly redirecting Students and Mentors to their respective dashboards.
- The complete collaboration workflow (Joining a room, chatting natively, checking off tasks) executes without interruption.
- The web browser interface responds correctly and remains responsive across different screen sizes.

6.3 RESULTS AND ANALYSIS

The intensive testing lifecycle showed that the Smart Collaboration and Career Guidance System (CollabSphere) handled data securely and performed successfully across all major interactive modules.

The functional results of testing indicated that:

- Role-based authentication dynamically protected API routes and UI views correctly.
- Study rooms functioned effectively as self-contained collaborative bubbles, accurately tracking localized chat and Kanban data.
- The RAG AI pipeline successfully avoided hallucination by properly prioritizing text from the localized knowledge base.
- Financial interactions involving Mentor Connections processed successfully, securely creating Razorpay orders and handling payment verifications flawlessly.
- Navigation between dashboard components (Find Mentor, Courses, AI Bot, User Profile) transitioned smoothly without triggering complete page reloads.

Overall, the final system easily met the expected functional requirements, demonstrating exceptional performance, cloud reliability, and modern usability during end-to-end testing.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 OUTPUT SCREENSHOTS

7.1.1 Secure Login and Registration Portal

The authentication screen serves as the secure entry point to the platform and provides a visually appealing, split-screen layout combining platform branding with a clean, modern login form. Users can securely sign in or register new accounts by providing their credentials (email and dynamically hashed passwords). The system utilizes robust JSON Web Token (JWT) based session management, ensuring that users remain securely logged in across page reloads while preventing unauthorized access to protected dashboard tabs. Role-based access control automatically redirects the user to either the Student Dashboard or Mentor Dashboard upon a successful login event, ensuring a personalized experience right from the start.

Fig. 7.1.1 User Authentication Interface

7.1.2 Centralized Study Room Hub

The centralized study room hub acts as the core collaborative ecosystem for peer-to-peer engagement. The student dashboard displays all currently active virtual study rooms using an intuitive grid format. Each room card clearly presents the subject topic, an optional descriptive summary, and live active member counts. Students can easily browse through different domains and click to request access to subject-specific collaborative spaces from this single, unified

interface. This eliminates the isolation of remote learning by transforming individual study plans into interactive group learning sessions, fostering a highly collaborative and motivating environment.

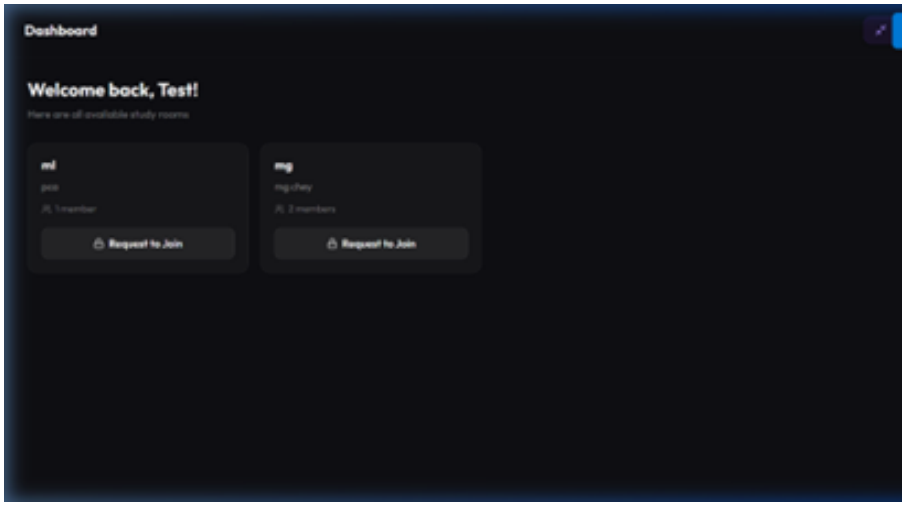

Figure 7.1.2: Centralized Study Room Hub

7.1.3 Mentor Exploration and Connection Interface

The 'Find Mentor' module effectively bridges the gap between academic preparation and real-world industry experience. It displays a comprehensive list of registered and verified professional mentors using dynamic profile cards. Each card highlights the mentor's specific domain of expertise, current job role, per-minute counseling pricing, and any available promotional discounts. Students can initiate a connection request directly from the card interface. This action triggers a secure backend approval workflow notifying the mentor. The interface is highly responsive, visually indicating the status of the connection (e.g., "Request Sent" or "Approved"), thereby streamlining the professional networking process.

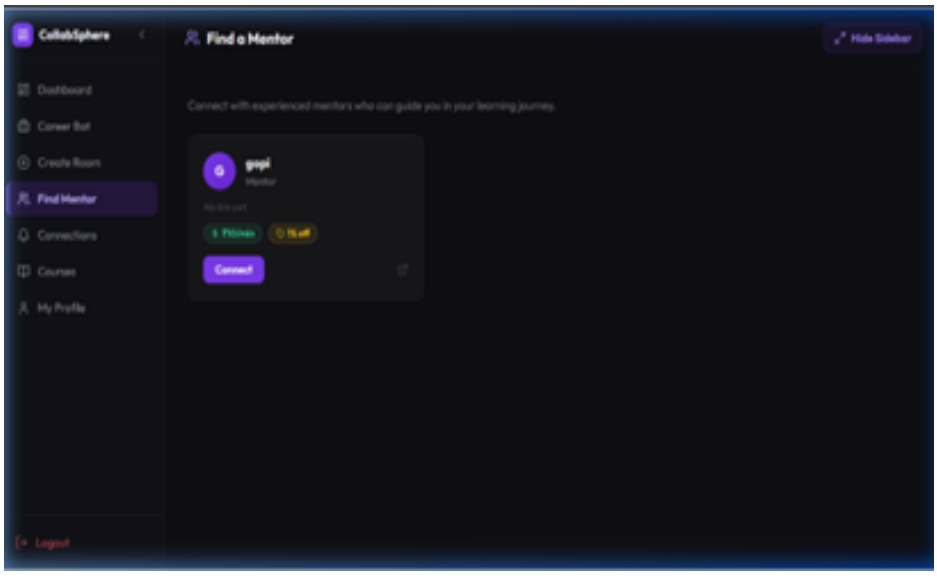

Figure 7.1.3: Mentor Discovery Module

7.1.4 Intelligent Career Navigator Chatbot

The Career Navigator acts as a personal, 24/7 virtual counselor for the student. The AI bot combines a specialized, curated career knowledge base with a powerful external Language Model API (e.g., Grok AI) to deliver highly context-aware responses. It accurately answers complex student queries regarding technical career paths, necessary skill development, and customized learning roadmaps. The chat interface is modern and user-friendly, supporting both free-form text input and quick-prompt suggestions (e.g., "How to become a Data Scientist?") for faster interaction. The bot's responses are dynamically formatted with markdown, rendering clear bullet points, bold text, and code snippets to maximize readability.

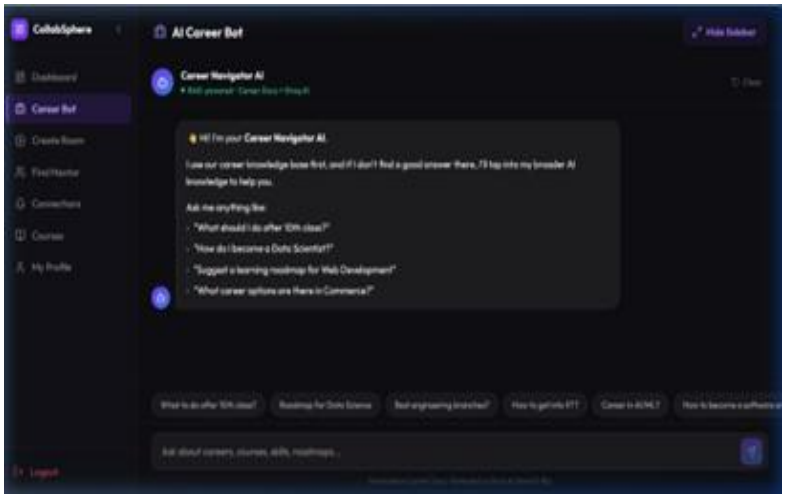

Fig. 7.1.4. AI Career Bot Interface

7.1.5 Collaborative Study Room Creation

The 'Create Room' panel empowers students to take leadership of their own learning by forming subject-based or project-based study groups. The interface provides a simple yet effective modal where the student enters an engaging room topic and an optional detailed description of the study goals. Once the form is submitted, the backend instantly provisions the room structure in the database, and it immediately becomes discoverable on the central hub. The creator is granted administrative rights over the room, allowing them to manage incoming peer requests. This feature guarantees that the platform remains highly dynamic and student-led.

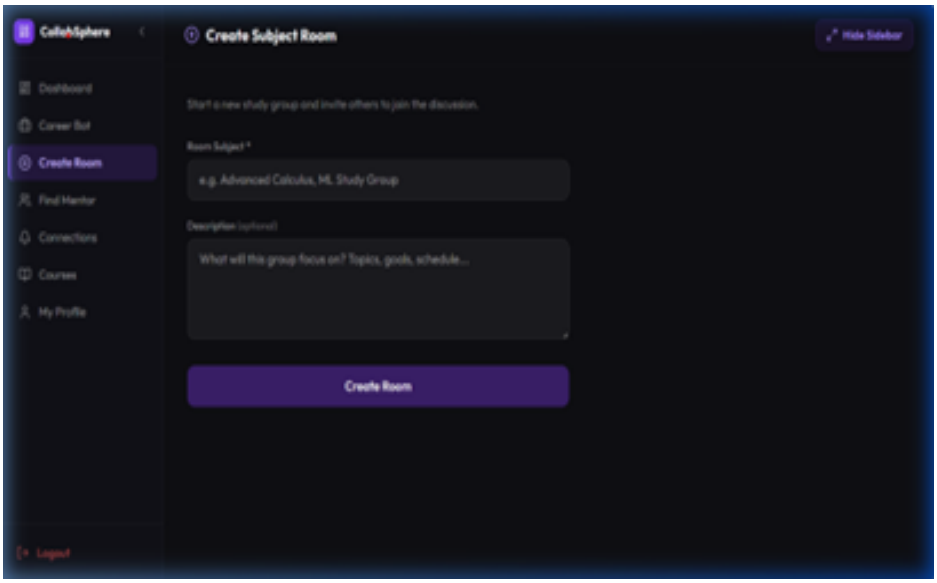

Figure 7.1.5: Create Study Room Panel

7.1.6 Secure Payment Integration for Mentorship

To ensure that industry professionals are compensated for their valuable time and expertise, a secure and seamless payment gateway is fully integrated into the mentorship workflow. Once a mentor approves a student's connection request, the student is prompted via a dedicated PaymentModal interface. The system securely processes the transaction using Razorpay (or a similar secure banking API webhook), ensuring that sensitive financial data is never directly exposed. Only upon successful payment confirmation does the platform fully unlock the direct communication channels, exclusive resources, and dedicated counseling benefits offered by the selected mentor.

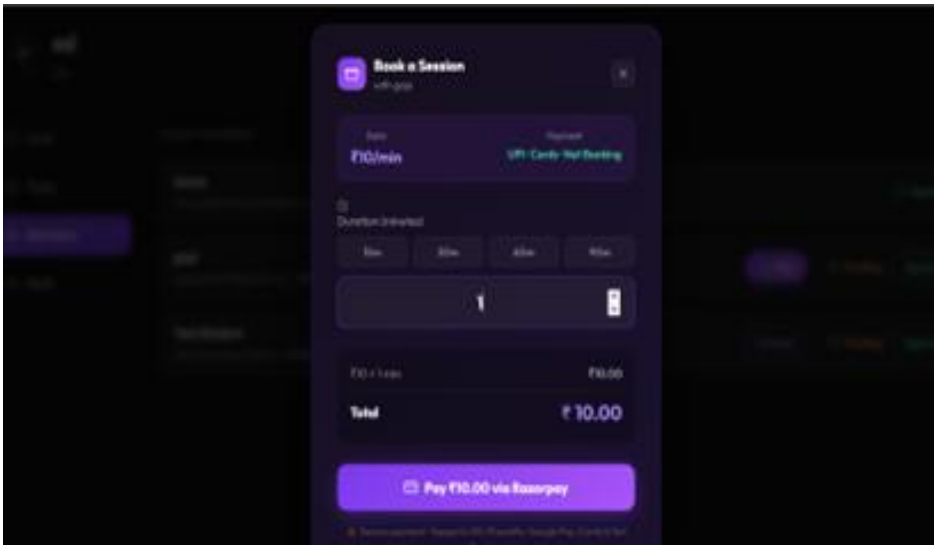

Figure 7.1.6: Secure Payment Modal using Razorpay

7.1.7 Real-Time Room Availability and Connectivity

To prevent students from joining dead or inactive spaces, the platform employs a real-time availability tracking system for its virtual study rooms. Students can instantly observe the current status of the room (Active, Full, or Archived) before attempting to join. The interface dynamically reflects the presence of connected peers, updating the active participant count as users enter or leave the room. This live-state functionality guarantees that students only invest their time in collaborative spaces that are currently populated and actively discussing the relevant career or academic topics.

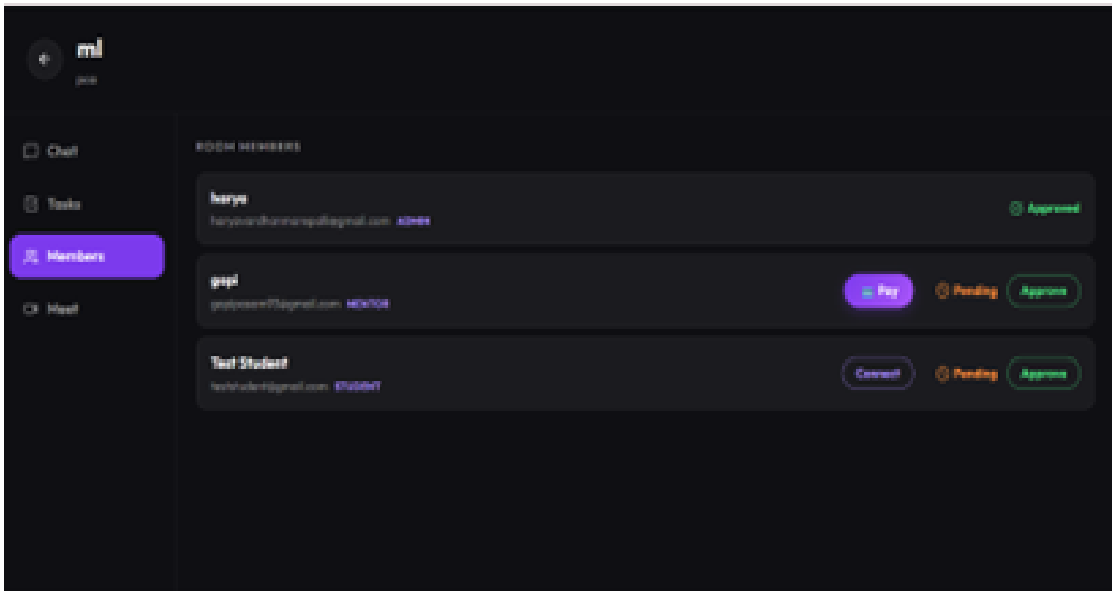

Figure 7.1.7: Room Availability and Interaction Tracking

7.2 PERFORMANCE ANALYSIS

The proposed Smart Collaboration and Career Guidance System (CollabSphere) demonstrated effective, highly reliable performance during implementation and testing. The Retrieval-Augmented Generation (RAG) pipeline successfully generated personalized, contextually accurate answers without sacrificing response speed. The system effectively handled concurrent study room states, ensuring that group chat messages and Kanban task updates reflected universally across member instances. Furthermore, external API workflows involving Razorpay behaved impeccably under localized conditions.

Performance observations include:

- Fast LLM response generation and seamless context retrieval from the local knowledge base.
- Reliable, real-time database updates for Kanban tasks and chat messaging.
- Effective load-balancing of dynamic React frontend components ensuring a jitter-free UI.
- Smooth frontend-backend interactions and secure transmission of JWT authentication blocks.
- Secure, error-free cryptographic verifications of Razorpay payment signatures and webhooks.

The results indicate that the system efficiently performs as an integrated academic super-app while maintaining excellent usability and responsiveness.

7.3 COMPARISON WITH EXISTING SYSTEM

To evaluate the effectiveness of the proposed system, a comparison was made between traditional, isolated study planning approaches (utilizing multiple disconnected applications) and the proposed CollabSphere architecture.

Table 7.1: Comparison of Traditional Platforms and Proposed

Metric	Traditional Disconnected Systems	Proposed CollabSphere System

Peer Collaboration	Fragmented across chat apps (WhatsApp) and separate task trackers (Trello)	Unified Study Rooms featuring persistent Chat alongside integrated Kanban tasks
Career Guidance	Generic, unlocalized, and primarily manual web searches	Automated, highly contextual RAG AI chatbot built on localized academic documentation
Mentor Accessibility	Extremely limited; reliant on informal social networking (LinkedIn)	Dedicated browsing directory connecting verified mentors to students
Payment Processing	Unstructured, non-transparent transfers (UPI) directly between parties	Fully integrated, escrow-based Razorpay mechanism with historical tracking
User Experience (UX)	High cognitive load due to constant app-switching	Streamlined, dedicated dashboards tailored specifically to Student and Mentor roles
Academic Efficiency	Manual coordination requires significant time from the student	Centralized platform radically improves organization and academic efficiency

The comparison clearly shows that the proposed CollabSphere system significantly improves collaboration efficiency, structured career planning, mentor transparency, and overall user personalization compared to traditional, static, and disjointed methodologies.

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 SUMMARY OF THE PROJECT

The proposed Smart Collaboration and Career Guidance System (CollabSphere) was developed to provide students and professional mentors with a unified, intelligent, and highly secure web-based platform for centralized academic management. Unlike traditional setups requiring disjointed applications for peer chats, task management, and mentor payouts, the system seamlessly integrates all these functional layers into one dedicated academic ecosystem.

The system integrates collaborative study rooms, Kanban-style task boards, advanced Retrieval-Augmented Generation (RAG) AI functionalities, and professional mentorship gateways to enhance holistic learning efficiency. By combining reliable career document retrieval, real-time peer communications, and structured Razorpay-backed mentor payments, the system transforms conventional academic discovery into an intelligent, protected networking framework.

Overall, the developed solution empowers students to make well-informed career transitions, significantly reduces collaboration bottlenecks, and constructs an equitable bridge for students worldwide to access high-quality professional mentorship securely.

8.2 ACHIEVEMENTS

The major achievements of the proposed Smart Collaboration and Career Guidance System include:

- Successful development of a unified peer collaboration and study room architecture.
- Seamless implementation of Kanban-style task tracking updating in real time.
- Integration of a bespoke Retrieval-Augmented Generation (RAG) AI Chatbot using the Groq Llama 3.1 LLM for accurate, context-aware career guidance.
- Development of segregated, highly interactive real-time analytic dashboards for both Students and Mentors using React 19.
- Successful implementation of secure JSON Web Token (JWT) role-based authentication and routing.

- Flawless integration of the Razorpay financial gateway—including Route payouts and escrow mechanics—securing mentorship interactions.
- Successful bridging of frontend interfaces, Flask backend server architectures, and relational PostgreSQL database models.

8.3 LIMITATIONS

Despite the high level of technical effectiveness achieved by the developed system, certain limitations exist within the scope of the current implementation:

- The collaborative room environment currently focuses on text-based group chat and lacks built-in native live video or audio conferencing capabilities.
- The application is exclusively deployed as a web-based dashboard and does not currently exist as a native mobile application (iOS/Android).
- The AI Career Guidance component fundamentally relies on continuous internet and API availability (Groq Cloud) to generate completions.
- Mentor connections are initiated manually via student browsing rather than being autonomously paired via an algorithmic AI-matching engine.
- Real-time browser Push Notifications or SMS alert systems are not currently supported; external alerts are handled purely via standard transactional emails.

8.4 FUTURE ENHANCEMENTS

Several robust enhancements can be incorporated into subsequent iterations of the proposed system to dramatically improve its scalability, enterprise value, and functional depth:

- Integration of WebRTC-based live video conferencing directly alongside the study room chat architectures.
- Development of a native cross-platform mobile application to enable native hardware push notifications and offline task inspection.
- Implementation of a machine learning-based matchmaking algorithm to auto-suggest mentors to students based on dynamic profile clustering and bio sentiment analysis.
- Introduction of an administrative/institutional dashboard for universities to monitor aggregated student engagement, learning trends, and mentor effectiveness.

- Addition of deep multi-language and regional support within the UI and the RAG document repository to service non-English speaking demographics better.
- Integration of calendar synchronization APIs (e.g., Google Calendar, Microsoft Outlook) to automate the scheduling loop for booked mentor sessions.

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